

Offer Annexes — Compliance Matrix, Exclusions & Assumptions

Power On BESS — 59 MW / 120 MWh, Plot 26, Psevdas, Larnaca, Cyprus

Client / Project: H.E.S.S. Hybrid Energy Storage Systems Ltd

Companion to: Initial Budgetary Offer (BESS / BoP / TS)

Date: 30 July 2026

Basis docs: BESS Scope of Supply · ANNEX-II · T14 · TSOC Connection Terms (Apr 2025) · PT Technical Information v2.0

Annex A — Technical Compliance Matrix

COMPLY meets requirement **EXCEEDS** better than required **CONFIRM** compliant, value to be verified/measured
OPEN to be fixed at design freeze

A.1 — BESS (PACKAGE A)

Requirement	Specified	Offered	Status
Max power at POC	50 MW	50 MW at POC (59 MW installed, 6 MV skids)	COMPLY
Guaranteed energy at POI (10 yr)	≥100 MWh	100 MWh held via 120.36 MWh oversize + Year-3 augmentation	COMPLY
Point of connection	132 kV	132 kV via new KYEA substation (Package B)	COMPLY
Nameplate DC (initial)	120 MWh	120.36 MWh — 24 × 5.015 MWh LFP containers	COMPLY
Cell chemistry	LFP, Tier-1	EVE 314 Ah LFP (Tier-1)	COMPLY
Discharge duration	1–3 h min	2.4 h at 50 MW / 120 MWh	COMPLY
Cycle life	≥6,000	8,000 cycles	EXCEEDS
Round-trip efficiency @POC	≥80%	≥80% at POC (all losses incl.); factory value to be confirmed	CONFIRM
Standby consumption	≤0.15·C/day	Aux ~25 kW; to be confirmed against 0.15·C limit at PoC	CONFIRM
Grid-code control (T14)	Sub-second P/Q; droop / FSM / LFSM-O/U; Q-V	Kehua BCS1250K-C-HUD — full grid-code functions; grid-forming (VSG) available in same firmware	COMPLY
Grid-forming (VSG)	Not explicitly mandated in ANNEX-II / T14	Available on offered PCS at no re-spec — enable if TSOC requires	CONFIRM
Fire protection (ANNEX-II §C)	Auto fire alarm + suppression; water supply	Pack/container aerosol + gas detection + DN65 water tie-in; site fire-water system in Package B	COMPLY
System independence (ANNEX-II §A)	No common LV busbars / aux transformers with HV substation	Independent EMS + separate auxiliaries; data exchange only via IEC 61850/104	COMPLY

Enclosure / corrosion	Coastal, outdoor	IP55, C5-M, liquid cooling, –30...+55 °C	COMPLY
Product warranty	≥3 years	10 years product (DC+AC) + performance ≥100 MWh @POI for 10 yr	EXCEEDS
Delivery	DDP Larnaka	DDP Larnaka / Psevdas	COMPLY

A.2 — 132/33 KV SUBSTATION & TS CONNECTION (PACKAGE B)

Requirement	Specified	Offered	Status
KYEA substation scope	Applicant-built full KYEA (Conn. Terms §5.1)	Full 132/33 kV substation EPC included	COMPLY
Main transformer T1	63 MVA, 132/33 kV, YNd11, ONAN/ONAF, OLTC, uk 21%, Tier 2	Assumed budget to compliant spec (PT v2.0); firmed on binding OEM quote	CONFIRM
33 kV insulation level	170 kV BIL / 70 kV AC (36 kV class)	170/70 per PT Technical Information v2.0	COMPLY
Earthing / auxiliary transformer T2	33 kV zig-zag, solid earthing (no NER), ≥20 kA/3 s	500 kVA zig-zag ZN, solid earthing — included	COMPLY
132 kV bay	CB, disconnectors, ES, surge arresters, CT/VT, cable termination	Included (AIS/GIS per TSOC-approved design)	COMPLY
33 kV switchgear	MV switchgear, eco-gas per TSOC	8 panels (6 feeders + transformer + bus coupler), eco-gas	COMPLY
Protection, control, RTU, metering, PQR	TSOC-approved; RTU/telecom to ECCC	Included; settings prepared by Lighthief, TSOC-approved before energisation; metering CT/VT supplied	COMPLY
CT arrangement	Per ISM bay protection SLD (6 vs 9 CT for REF)	6-CT base priced; 9-CT option — fix at TSOC-approved SLD	OPEN
Earthing system	T14 study + TSOC acceptance; touch/step voltages	Design, soil resistivity, measurements + TSOC acceptance included	COMPLY
Fire protection (ANNEX-II §C.IV.7)	≥20 m³ fire-water / permanent supply	20 m³ tank + electric/diesel pumps + hydrants + hose reels	COMPLY
HV safety coordination	TSOC HV Authorization (T12.4.2.2)	Lighthief Safety Coordinator + RAMS included	COMPLY
Seismic design	CYS EN 1998-1	Designed to Zone III, agR 0.25 g (zone/agR to be confirmed vs Zone II 0.23 g)	CONFIRM
33 kV cable	Per SLD (feeders, cross-section)	6 feeders ≤300 m, Cu 630 mm² (optimisable to Al 630 / Cu 300)	CONFIRM

Open / confirm items do not affect the safe, compliant delivery of the plant — they are values fixed at design freeze (TSOC-approved SLD, seismic zone, RTE/standby measured values) or optimisation choices (33 kV cable cross-section). Full clause-by-clause compliance to Annex A (RAAEY) is provided on award.

Annex B — Exclusions & Assumptions

B.1 — EXCLUSIONS (NOT IN THE OFFERED PRICES)

- VAT (Cyprus 19%, reclaimable), import duties, taxes, connection/permit/authority fees and land acquisition.
- 132 kV cable from the existing Psevdas TS to the KYEA, and any Psevdas TS extension/reinforcement — ISM scope under the connection terms.
- Installation and programming of main and check energy meters (ISM/ISD scope); metering CT/VT are included.
- Ongoing telecom link fees to ECCC/TSOC after handover.
- Long-term service agreement / extended warranty / spare parts beyond the quoted 10-year LTSA option.
- Any change arising from TSOC or authority requirements after the date of issue (subject to written variation).
- Works, permits or utilities not explicitly listed in the offer scope.

B.2 — ASSUMPTIONS (BASIS OF THE ASSUMED BUDGETS)

- **Configuration:** 24 containers + 6 MV skids (initial); Year-3 augmentation (+5 containers, +2 skids) to hold ≥ 100 MWh at POI — per Linyang quotation of 2 July 2026.
- **Grid support mode:** priced on standard grid-code functions; grid-forming available on the same PCS with no equipment re-spec — to be confirmed with TSOC.
- **Transformers T1 & T2:** assumed budget to the compliant PT v2.0 spec (170/70); firmed on binding manufacturer quotation. A surplus-unit value-engineering option may be offered separately subject to TSOC acceptance of deviations.
- **33 kV switchgear, 132 kV bay, civil works:** budgetary allowances pending supplier quotations, the TSOC-approved SLD (CT count) and the local civil bill of quantities.
- **33 kV cable:** 6 feeders up to 300 m, Cu 630 mm²; cross-section may be optimised with the design.
- **Contracting:** Package A (BESS) + BoP via Lighthief Cyprus Ltd; Package B (TS) via Lighthief Farmy Fotowoltaiczne Sp. z o.o. (coordinated contracts).
- **Fire-water system** (20 m³ tank, pumps, hydrants, hose reels) is in the TS package; the DN65 container tie-ins are in the BoP package.
- **Commercial:** EUR, ex-VAT; validity 30 days; payment per Lighthief standard EPC milestones; 5% performance bond.
- **Schedule:** BESS T0 + 90 days production + ~75 days transit (DDP); main transformer targeted on site Q1 2027; energisation subject to TSOC hold points. Linyang pricing assumes shipment by end-December 2026.
- **Site data:** altitude 264 m; ambient design 45 °C; pollution Class III; seismic Zone III, agR 0.25 g (to be confirmed).

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Confidential — annexes to the Initial Budgetary Offer for H.E.S.S. Hybrid Energy Storage Systems Ltd, 30 July 2026. Read together with the offers; subject to the assumptions above.